Trigonometric Functions & the Unit Circle

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A terminal ray in standard position passes through the point $P(-8, 15)$. Find the exact values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Step 1: Compute the radial distance $r = \sqrt{x^2 + y^2}$: $r = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. Step 2: Apply circular trigonometric definitions: $\sin\boldsymbol{\theta} = \frac{y}{r} = \frac{15}{17}$, $\cos\boldsymbol{\theta} = \frac{x}{r} = -\frac{8}{17}$, $\tan\boldsymbol{\theta} = \frac{y}{x} = -\frac{15}{8}$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Given vectors $\mathbf{u} = \langle 4, -3\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.

(a) Linear combination: $2\langle 4, -3\rangle - 3\langle -1, 5\rangle = \langle 8 + 3, -6 - 15\rangle = \boldsymbol{\langle 11, -21\rangle}$. (b) Magnitude and unit vector: $||\mathbf{u}|| = \sqrt{4^2 + (-3)^2} = \mathbf{5}$, $\hat{\mathbf{u}} = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$. (c) Dot product: $\mathbf{u} \cdot \mathbf{v} = (4)(-1) + (-3)(5) = -4 - 15 = \mathbf{-19}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

State the amplitude, period, phase shift, and midline of $f(x) = 3\sin(2x - \pi) + 1$.

Step 1: Factor the argument into standard sinusoidal form $f(x) = A\sin(B(x - C)) + D$: $f(x) = 3\sin\left(2\left(x - \frac{\pi}{2}\right)\right) + 1$. Step 2: Read off the four sinusoidal invariants:Amplitude: $|A| = |3| = \mathbf{3}$ .Period: $T = \frac{2\pi}{|B|} = \frac{2\pi}{2} = \boldsymbol{\pi}$.Phase Shift: $C = \boldsymbol{\frac{\pi}{2}}$ units to the right.Midline: Horizontal line $\mathbf{y = 1}$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A population doubles every 12 years, starting from $P_0 = 500$. When does it reach 4000?

Step 1: Formulate the exponential doubling model: $P(t) = P_0 \cdot 2^{t/12} = 500 \cdot 2^{t/12}$. Step 2: Set $P(t) = 4000$ and isolate the exponential base: $500 \cdot 2^{t/12} = 4000 \Rightarrow 2^{t/12} = \frac{4000}{500} = 8$. Step 3: Equate exponents of base 2: $2^{t/12} = 2^3 \Rightarrow \frac{t}{12} = 3 \Rightarrow t = \mathbf{36}$ years.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

Step 1: Multiply both sides by r: $r^2 = 6r\cos\theta - 8r\sin\theta$. Step 2: Substitute $r^2 = x^2 + y^2$, $r\cos\theta = x$, and $r\sin\theta = y$: $x^2 + y^2 = 6x - 8y \Rightarrow (x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 = 25$. $\boldsymbol{(x - 3)^2 + (y + 4)^2 = 25}$. The curve is a circle with center $\boldsymbol{(3, -4)}$ and radius $\mathbf{R = 5}$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.

Step 1: Isolate $\cos t$ and $\sin t$: $\cos t = \frac{x-3}{4}$, $\sin t = \frac{y+2}{5}$. Step 2: Apply the Pythagorean identity $\cos^2 t + \sin^2 t = 1$: $\mathbf{\frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1}$. This is a vertical ellipse centered at $\mathbf{(3, -2)}$ with major-axis vertices at $\mathbf{(3, -7)}$ and $\mathbf{(3, 3)}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle moves along $x(t) = 2t + 1$, $y(t) = t^2 - 3$. Find the Cartesian equation of the path.

Step 1: Solve the horizontal parametric equation for parameter t: $x = 2t + 1 \Rightarrow 2t = x - 1 \Rightarrow t = \frac{x-1}{2}$. Step 2: Substitute t into the vertical equation $y(t)$: $y = \left(\frac{x-1}{2}\right)^2 - 3 = \mathbf{\frac{(x-1)^2}{4} - 3}$. The trajectory is an upward-opening parabola with vertex at $(1, -3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the transformation grid below. If the original point A is mapped to A', what is the vector representing this translation?

The vector representing a translation from point $P = (x_1, y_1)$ to point $P' = (x_2, y_2)$ is given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, point A is at $(1, 2)$ and point A' is at $(-5, 2)$. Therefore, the translation vector is $\langle -5 - 1, 2 - 2 \rangle = \langle -6, 0 \rangle$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedFind all values of θ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.

Step 1 (Find Pole Crossings where r = 0): $4\sin(3\theta) = 0 \Rightarrow \sin(3\theta) = 0$. For $\theta \in [0, 2\pi)$, the argument $3\theta \in [0, 6\pi)$. The sine function equals zero at integer multiples of π: $3\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$. $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$. Step 2 (Find Maximum Distance from Pole): Distance from pole is $|r| = |4\sin(3\theta)|$. The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum distance from t.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Unit 3 · Trigonometric and Polar Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · AdvancedFind the matrix R that rotates vectors by $\theta = \frac{2\pi}{3}$ radians counterclockwise, and use it to find the image of the vector $\vec{w} = (4;\ -2)$. Verify that the magnitude of $\vec{w}$ is preserved under the transformation.

Step 1 (Formulate Rotation Matrix): For $\theta = \frac{2\pi}{3}$: $\cos(2\pi/3) = -\frac{1}{2}$ and $\sin(2\pi/3) = \frac{\sqrt{3}}{2}$: $R = \left(-1/2,\ -\sqrt{3}/2; \sqrt{3}/2,\ -1/2\right)$. Step 2 (Transform $\vec{w}$): $R\vec{w} = \left(-1/2,\ -\sqrt{3}/2; \sqrt{3}/2,\ -1/2\right)(4;\ -2) = \left((-1/2)(4) + (-\sqrt{3}/2)(-2); (\sqrt{3}/2)(4) + (-1/2)(-2)\right) = \left(-2 + \sqrt{3}; 2\sqrt{3} + 1\right)$. Step 3 (Verify Magnitude Preservation): $|\vec{w}|^2 = 4^2 + (-2)^2 = 16 + 4 = 20$. .

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 2 · MediumVector $\vec{u} = \langle 4,\ -3 \rangle$ and vector $\vec{v} = \langle 1, 7 \rangle$. Calculate the magnitude of each vector, determine the dot product $\vec{u} \cdot \vec{v}$, and find the exact angle θ between them in radians.

Magnitudes: $|\vec{u}| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. $|\vec{v}| = \sqrt{1^2 + 7^2} = \sqrt{1 + 49} = \sqrt{50} = 5\sqrt{2}$. Dot Product: $\vec{u} \cdot \vec{v} = (4)(1) + (-3)(7) = 4 - 21 = -17$. Angle Between Vectors: Use $\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$: $\cos\theta = \frac{-17}{(5)(5\sqrt{2})} = \frac{-17}{25\sqrt{2}} = \frac{-17\sqrt{2}}{50} \approx -0.48083$. .

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve the trigonometric equation $2\cos^2 x - \sin x - 1 = 0$ for all x in the interval $[0, 2\pi)$.

Step 1: Apply the Pythagorean identity $\cos^2 x = 1 - \sin^2 x$: $2(1 - \sin^2 x) - \sin x - 1 = 0 \Rightarrow 2\sin^2 x + \sin x - 1 = 0$. Step 2: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x + 1) = 0 \Rightarrow \sin x = \frac{1}{2}$ or $\sin x = -1$. Step 3: Extract all solutions in $[0, 2\pi)$: $\mathbf{x \in \left\{\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}\right\}}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Prove the trigonometric identity $\frac{\sin(2x)}{1 + \cos(2x)} = \tan x$ and state the domain exclusions.

Step 1: Apply double-angle identities $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = 2\cos^2 x - 1$: LHS $= \frac{2\sin x\cos x}{1 + (2\cos^2 x - 1)} = \frac{2\sin x\cos x}{2\cos^2 x}$. Step 2: Cancel $2\cos x$ (for $\cos x \neq 0$): $\frac{\sin x}{\cos x} = \tan x = \text{RHS}$. ■ Domain exclusions: $\mathbf{x \neq \frac{\pi}{2} + k\pi}$ for $k \in \mathbb{Z}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Level 3 · AdvancedAn invasive plant covers an area modeled by $A(t) = 120(1.15)^t$ square meters, where t is measured in weeks. Write the equation setup to determine when the area reaches 500 square meters, and find the value of t to three decimal places with correct units.

Setup: Write the equation before numerical solving: $120(1.15)^t = 500 \Rightarrow (1.15)^t = \frac{500}{120} = \frac{25}{6}$. Solve: Take the natural logarithm: $t\ln(1.15) = \ln\left(\frac{25}{6}\right) \Rightarrow t = \frac{\ln(25/6)}{\ln(1.15)}$. Evaluating on the calculator: $t \approx \frac{1.427116}{0.139762} \approx 10.211$ weeks. Final answer: 10.211 weeks.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Rates of Change & Concavity

[AP Precalculus Exam Format & Modeling Framework · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

[Unit 4 · Parameters, Vectors, and Matrices · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$: $= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x =$ RHS. ✓

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Inverse & Composite Functions

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Angle $\frac{5\pi}{6}$ lies in Quadrant II with reference angle $\frac{\pi}{6}$. Therefore, $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ and $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$, giving coordinates $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

a. 315° ✓

b. 225°

c. 300°

d. 135°

Multiply by $\frac{180^\circ}{\pi}$: $\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 7 \times 45^\circ = 315^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the terminal ray of angle α passes through the point $(-3, -4)$, what is $\sin\alpha$?

a. $-\frac{4}{5}$ ✓

b. $-\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{3}{5}$

Radius $r = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9 + 16} = 5$. Then $\sin\alpha = \frac{y}{r} = -\frac{4}{5}$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

a. $\sqrt{2}$ ✓

b. $-\sqrt{2}$

c. $\frac{\sqrt{2}}{2}$

d. 2

Cosine is an even function: $\cos(-\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$. Thus $\sec(-\pi/4) = \frac{1}{\cos(\pi/4)} = \frac{2}{\sqrt{2}} = \sqrt{2}$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

a. **Amplitude** $= 4$**, Period** $= \frac{2\pi}{3}$ ✓

b. Amplitude $= -4$, Period $= 3$

c. Amplitude $= 4$, Period $= 3\pi$

d. Amplitude $= 2$, Period $= \frac{2\pi}{3}$

Amplitude is $|A| = |-4| = 4$. For $B = 3$, the period is $\frac{2\pi}{B} = \frac{2\pi}{3}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A Ferris wheel of radius 25 m completes one rotation every 60 seconds. Passengers board at the bottom, 2 m above the ground. Which cosine equation models height $h(t)$ in meters?

a. $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$ ✓

b. $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

c. $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

d. $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

Centerline is $d = 2 + 25 = 27$. Amplitude is 25. Period is $60 \Rightarrow B = \frac{2\pi}{60} = \frac{\pi}{30}$. Since it starts at minimum height 2, an inverted cosine is used: $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

a. $\frac{\pi}{8}$ **units to the left** ✓

b. $\frac{\pi}{8}$ units to the right

c. $\frac{\pi}{2}$ units to the left

d. $\frac{\pi}{4}$ units to the left

Factor out 4: $g(x) = 2\sin\left(4\left(x + \frac{\pi}{8}\right)\right)$. The phase shift is $\frac{\pi}{8}$ units to the left.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\csc x$

d. $\cot x$

Rewrite in terms of sine and cosine: $\frac{\sin x/\cos x}{1/\cos x} = \frac{\sin x}{\cos x} \cdot \cos x = \sin x$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which expression is equivalent to $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all defined θ?

a. 1 ✓

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

Use Pythagorean identities: $1 - \cos^2\theta = \sin^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$. Therefore, $\sin^2\theta \cdot \csc^2\theta = \sin^2\theta \cdot \frac{1}{\sin^2\theta} = 1$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 5. Find the arc length subtended by a central angle of $\frac{1\pi}{2}$ radians.

a. $5/4\pi$

b. $5/2\pi$ ✓

c. $7/2\pi$

d. 5π

$s = r\theta = 5 \cdot \frac{1\pi}{2} = 5/2\pi$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Eliminate the parameter: $x = 3t$, $y = 3t$.

a. $y = 1x$ ✓

b. $y = 1x + 1$

c. $y = 2x$

d. $y = -1x$

Solve $x = 3t$ for t: $t = \frac{x}{3}$. Substitute into y: $y = 1x$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let $f(x) = 3x + 2$ and $g(x) = 1/3x - 2/3$. Are f and g inverse functions?

a. No, f and g are not inverses

b. **Yes, f and g are inverses** ✓

c. Cannot be determined

d. Only if $x = 0$

Check $f(g(x))$: this simplifies to x, so they are inverses.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

End behaviour of $f(x) = -2x^5$ as $x \to +\infty$:

a. $+\infty$

b. $-\infty$ ✓

c. 0

d. 2

Odd degree, negative leading coefficient $\to -\infty$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the determinant of the 2×2 coefficient matrix $A = (4, 2; 3, 4)$.

a. 22

b. **10** ✓

c. 13

d. 6

$\det(A) = ad - bc = (4)(4) - (2)(3) = 16 - 6 = 10$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which transformation shifts $f(x)$ right by 3?

a. $f(x + 3)$

b. $f(x - 3)$ ✓

c. $f(x) + 3$

d. $f(x) - 3$

Horizontal shift right: replace x with $x - 3$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the vector **v** directed from the point $P = (2, 5)$ to the point $Q = (7, 1)$. What is the component form of **v**?

a. $\langle -5, -4 \rangle$

b. $\langle 5, 4 \rangle$

c. $\langle 5, -4 \rangle$ ✓

d. $\langle -5, 4 \rangle$

The component form of a vector directed from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 5)$ and $Q = (7, 1)$. Therefore, the component form of **v** is $\langle 7 - 2, 1 - 5 \rangle = \langle 5, -4 \rangle$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.03 Parametric Equations & Modeling

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.03 Rates of Change & Concavity

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. **3** ✓

b. 2

c. 6

d. 1

We solve the problem by synthesizing harmonic form (linear combination of sine and cosine), angle-doubling relationships, and quadratic optimization on a restricted domain. **Step 1: Express the single-angle terms in harmonic form.** Recall the linear combination identity $A\cos x + B\sin x = R\cos(x - \alpha)$. For the terms $2\sqrt{3}\cos x - 2\sin x$: $2\sqrt{3}\cos x - 2\sin x = 4\left(\frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x\right)$ Using the cosine addition formula.

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